UNIT 10 APPROACHES TO BALANCE OF PAYMENTS *

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10.0 OBJECTIVES

After studying this Unit, you should be able to:

- identify different approaches to BoP;
- explain the main ingredients of these approaches;
- distinguish between monetary approach under fixed and flexible exchange rates;
- point out the main criticisms levelled against the monetary approach;
- make a synthesis of different approaches; and
- highlight the importance of these approaches in making BoP policy.

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* *Contributed by* Prof. S.K. Singh, Retd. Prof. IGNOU

10.1 INTRODUCTION

Some of the problems which policymakers generally face while analyzing the balance of payments are related to the disequilibrium deficit or surplus and their causes. Disequilibrium in the balance of payments is a general economic phenomenon in most developing countries. In this context, it is important to know the role of a country's central bank and the policy options it has for maintaining equilibrium in the balance of payments. Many theories in the field of international economics explain the balance of payments disequilibrium and the movements of international reserves in one way or another. Mainly there are four approaches. In the elasticities and absorption approaches, the focus is on the trade balance with unemployed resources. These approaches are associated with *Keynesian* theory. The distinctive feature of the monetary approach to the balance of payments is that it examines the implications for the stock equilibrium of the continuing flow of financial assets required to finance a continuing balance of payments surplus or deficit. A central proposition of the monetary approach to the BoP is that the balance of payments is a monetary phenomenon requiring analysis with the tools of monetary theory. This unit reviews the major approaches to BoP adjustments.

10.2 APPROACHES TO BoP

The alternative approaches to BoP adjustment are explained in this section. These approaches are as under:

i) The Elasticity Approach;

ii) The Absorption Approach;

iii) The Keynesian approach; and

iv) The monetary approach.

Let us discuss all the approaches one by one.

10.2.1 Elasticity approach

We begin first by considering the elasticity approach. The elasticities approach is based on the premise that relative prices determine exports and imports. In this way elasticities approach argues that the number of exports and imports will depend upon domestic prices, foreign prices and the exchange rate. This approach applies the *Marshallian* analysis of elasticities of supply and demand for individual commodities to the analysis of exports and imports as a whole. It is generally assumed that exports depend on the price of exports, and imports depend on the price of imports. These relations are then translated into elasticities.

The effect of devaluation on the trade balance depends on four elasticities:

a) the foreign elasticity of demand for exports;

b) the home elasticity of supply;

c) the foreign elasticity of supply of imports; and

d) the home elasticity of demand for imports.

The immediate effect of devaluation is a change in relative prices. If a country devalues by, for example, 20 percent, it means that import prices increase by 20 percent counted in home prices. An increase in import prices leads to a fall in the demand for imports. At the same time, import-competing industries will be in a better competitive situation. Exporters will receive 20 percent more in the home currency for every unit of foreign currency they earn. They can, therefore, lower their prices counted in foreign currency and will become more competitive. How much they can expand sales abroad depends primarily on the foreign demand elasticities for their goods.

The larger the elasticity, the greater the fall in the volume of imports. The value of the demand elasticity of imports depends, of course, on what types of goods the devaluing country imports. When the exporters, because of devaluation receive more for every unit of foreign currency they earn, they can lower their prices quoted in foreign currency. When they lower their prices they should be able to sell more. How much the quantity exported increased depends on the demand elasticity confronting the country's exporters.

The essence of the theory is embodied in the famous ***Marshall Lerner condition***: an improvement in competitiveness will improve the balance of payments if and only if the sum of the price elasticities of demand for imports by residents and exports by non-residents exceeds unity. The crucial feature of the condition is that it is the *sum* of the elasticities which must exceed one, not each elasticity separately. Hence both exports and imports can be price inelastic but for the condition to be satisfied. In effect, the exchange rate clears the balance of payments. A criterion for a change of the balance of trade in the desired direction can be established, assuming that export and import prices adjust to equate the demand for and supply of exports and imports.

The effect on the price level depends primarily on the economic policy accompanying devaluation. The inflationary impact could be limited if a tight monetary and fiscal policy is pursued jointly with devaluation. Another consideration to take into account is the effect of devaluation on income distribution. Again devaluation should result in a reallocation of resources away from the sector producing non-traded goods and into exports and import-competing sectors. In general, we can say that the factors of production employed in the export and import-competing sectors will benefit from devaluation.

The *Marshall-Lerner condition* is built on some drastic simplifications. It roughly assumes that the supply elasticities are large and that the trade balance is in equilibrium when devaluation takes place. The sort of mechanism ignored by Marshall-Lerner includes such processes as:

i) a change in relative prices increases exports;

ii) the increase in exports generates an increase in income;

iii) the higher level of income leads to more imports.

This latter increase in imports can be such that there is no improvement in the balance of payments despite the rise in exports. However, none of these assumptions invalidates the spirit of the *Marshall-Learner* condition, which says that the larger the respective demand elasticities, the more favourable the effect of devaluation on the trade balance. So long as the sum of elasticities exceeds unity, then an improvement in competitiveness will improve the balance of payments i.e., the *Marshall Lerner* condition.

The significance of Marshall-Lerner condition: Opinion is divided about the relevance of Marshall-Lerner, or competitiveness, to the balance of payments. On the one hand, some economists believe that competitiveness is a major determinant of the balance of payments. On the other hand, according to *Johnson* "it should be emphasized that the analysis of the effects of devaluation is completely independent of any critical magnitude condition applying to the elasticities of international demand. The relevant stability condition is monetary theoretic. The familiar elasticity condition (sum-of-the-elasticities-of demand–greater-than-unity) is completely irrelevant".

However, it should be noted that the elasticities approach in general can be relevant to the analysis of inflation and unemployment. The income effect, whose size does depend in part on *Marshall-Lerner* analysis, is not an unwelcome by-product but the object of the policy. Similarly, exogenous change in competitiveness e.g. by income policy, as a means to reduce unemployment is important in policymaking. The major flaw with the elasticities approach is that it only looked at the current account balance, not the overall balance of payments.

10.2.2 The Absorption Approach

Sydney Alexander of the IMF first presented the absorption approach in 1952. He sought to look at the balance of trade from the point of view of national income accounting:

Y = domestic production of goods and services

E = domestic absorption of goods and services, or domestic total expenditure

BT = balance of trade

BT = Y - E

The above identity is useful in pointing out that an improvement in the balance of trade calls for an increase in production relative to absorption. The absorption approach can be said to work only in the presence of unemployed resources. This approach argued that questions regarding the effects of changes in economic variables on the balance of payments were best assessed by ascertaining the effects on output and absorption. It concentrates on the relationships between real expenditure and real income and the relationships between these price levels. Absorption here is the name given to the aggregate of domestic demand (C + I + G), which is the number of goods and services taken off the market domestically. To *Alexander*, the foreign balance '**B**' is the difference between the total output of goods and services '**Y**' and the total absorption '**A**' of these goods and services by the home economy. Thus, B =Y–A.

If the total output is larger than the total expenditure, the country will have a surplus in its BoP and if the total expenditure is larger than the total output the country will have a deficit, and if output equals expenditure, the BoP will be in equilibrium. If a country has a deficit it can, in principle, close the deficit in one of two ways; by reducing expenditure or by increasing output. It is often difficult to increase output in the short run especially if the country already has full employment. Therefore, the chief means for reducing a deficit is usually an expenditure-reducing or switching policy. A tighter *monetary policy and fiscal policy* can be viewed as an efficient means of implementing an expenditure-reducing policy. If the country has a deficit in the BoP, it can pursue a tighter monetary and fiscal policy. Therewill be a deflationary effect on the national income and lead to a fall in imports, or at last act as a brake on the increase in imports. It will also have a positive effect on exports and import-competing industries.

The criticisms against the absorption approach: The following are some of the weaknesses of this approach:

i) *Machlup* has observed that certain elasticity considerations in the model had been ignored. *Alexander* had seen these situations in terms of different supply and demand conditions for commodities, which necessarily involved elasticities. However, *Alexander's* analysis played down the role of elasticity in his model.

ii) The estimation required to bring the absorption approach up to *the 'operational'* level was just as formidable as those involved in the elasticities approach. In the case of the absorption approach the relevant propensities had to be known before it was possible to say whether devaluation would be beneficial to the trade balance.

iii) It was necessary to embody within the model the fact that an improvement in the balance of trade of a devaluing country involved opposite changes in the trade balance of the rest of the world.

iv) This approach suffered from the difficulty that it was not suited for examining changes that only indirectly affected output and absorption: changes such as devaluation or inflation.

v) An important flaw and one shared with the elasticities approach is that it only looked at the current account balance and not at the overall balance of payments.

The result of these criticisms of the absorption approach was a reformulation which included elasticity effects and was more nearly a synthesis of the elasticity and the income approaches. It is important to realise how neatly the absorption model complements the elasticities approach. The traditional approach ignored supply-side effects and income effects. The absorption approach looks only at these two effects. The two approaches can be combined. *In brief*, the absorption approach focuses on the key factors omitted from the traditional approach and provides a general framework of analysis. Its single most virtue stems from its very existence, the idea that the balance of payments is a macroeconomic variable.

10.2.3 Keynesian Approach

The *Keynesian* balance of payments analysis is a coherent presentation of the macroeconomic case for import controls and devaluation. The *Keynesian* balance of payments theory was the absorption model of Alexander, of the IMF, in the early 1950s. As in all *Keynesian* models, the balance of payments on the current account, is analysed as a macroeconomic phenomenon in the goods market. The current balance of payments will necessarily equal the difference between aggregate domestic output and aggregate domestic expenditure (with a surplus if the output is large and vice versa). This conclusion follows from a manipulation of the basic national income; output '**O**' and expenditure '**E**', where O=E.

Although the part played by income changes in BoP adjustment is Keynesian in approach and method, Keynes himself took no direct part in its formulation or development. The Keynesian approach made possible the formulation of a new approach to the adjustment process by *Mrs Joan Robinson, R.F. Harrod, Fritz Machlup and others*. This new approach showed that an imbalance in BoP involved an adjustment in income, employment and output irrespective of what changes took place in prices and of how the deficit was financed. It showed an interaction between the BoP and national income which was a dual one: the '*adjustment process*', by which a BoP is adjusted by changes in the income levels of the home country and the rest of the world; and the '*transmission process*', which shows how variations in the national income of one country may through their balances of payments cause variations in the national incomes of other countries.

Thus, with the development of the income approach to BoP adjustment, a marriage was effected between the theory of International trade and business cycle theory from which much could be expected. It explains not only the adjustment process but also the transmission process. It shows that since deficits are adjustable by relative movements of national income, by reduction of income in the deficit country and increase of income in the surplus country, correction of a deficit must under static productivity conditions, necessarily involve a reduction in national income. The full condition of equilibrium for a country was now redefined as one in which total external receipts equalled total external payments at full employment. This linkage of external equilibrium for a country with its level of income and employment by the new approach enabled a much clearer and most realistic view to be taken of BoP policy in modern times.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What is meant by the Marshall-Lerner condition?

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2) Explain the main features of the absorption approach.

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3) The main merit of the Keynesian approach lies in integrating national income with BoP.

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4) What are the distinctive features of elasticity, absorption, and the Keynesian approach to BoP?

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5) How do these three approaches differ?

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10.2.4 The Monetary Approach

In this section, we examine the monetary approach to the balance of payments.

Evolution of monetary approach: *Alan Barrett* is of the view that some controversy surrounds the early origins of the monetary approach to BoP. Essentially the controversy is over whether *David Hume or Richard Cantillon* is the true forefather of the approach. *Ludwig von Mises* can be considered a forerunner of the monetary approach because he analysed the causes and cures of a persistent balance-of-payments disequilibrium. *Mises's* contribution to balance-of-payments analysis should be hailed as a doctrinal milestone in the development of the monetary approach and, much more importantly, as a shining exemplar of methodological individualism in monetary theory.

The monetary approach was developed in the 1950s and 1960s by the International Monetary Fund's research department under *Jacques J. Polak, and by Harry G. Johnson, Robert A. Mundell*, and their students at the University of Chicago and became fully developed during the 1970s. The monetary approach represents an extension of domestic monetarism (stemming from the Chicago school) to the international economy in that it views the balance of payments as an *essentially monetary phenomenon*. That is, money plays a crucial role in payments.

Mundell included the capital account of BoP in the discussion, and incorporated two policy instruments. One was the growing reluctance of countries to use devaluation as a policy instrument, and the other was the growing constraints on the use of trade barriers arising from G.A.T.T negotiations. While it was also *Mundell* who recognized that, in a model of capital mobility, the central bank controls not the money supply and employment but domestic credit and the balance of payments, Harry Johnson popularized the concept of the fundamentally monetary nature of the balance of payments. *Jacques Polak's* approach to monetary management in the context of the balance of payments disequilibrium evolved in the 1950s in work concerned with Latin American economies. The monetary approach to the balance of payments is an extension of closed economy monetary theory, stressing the stability of the demand for money function and considering the channels through which changes in the money supply, out of line with changes in money demand, affect the economy.

10.2.4.1 Tenets of monetary approach

To start with let us understand the basic assumptions of this approach which are as under:

i) Assumes flexible exchange rates. In other words, the currencies in question are traded in foreign exchange markets with minimal or no government intervention.

ii) It focuses exclusively on the behaviour of international investors trying to decide between securities denominated in two different currencies. Based on this the monetary approach is also called the *asset approach to exchange rate determination*.

iii) The approach emphasizes the relevance of the changes in monetary policy and the resulting changes in real returns on securities denominated in different currencies. After investors observe these changes in real returns, they express their preference for security, which leads to buying or selling certain currencies and, therefore, changes in the exchange rate.

iv) The demand for money is a stock demand and is a stable function of income, prices, wealth and interest rate.

v) The supply of money is a multiple of the monetary base, including domestic credit and the country's foreign exchange reserves.

vi) The demand for nominal money balances is a positive function of nominal income.

Given these assumptions, the BoP is regarded as a monetary phenomenon. The very concept of a BoP implies the existence of money. The proponents of the monetary approach regard the balance of payments as being fundamentally a monetary phenomenon. An analysis which does not place money at the centre of attention is seen as inferior. For them, the attraction of the monetary approach is that it analyses the relationships directly relevant to the money account and not in terms of other behavioural relationships which are only indirectly relevant to the money account. Considerations of this relationship can lead us to the conclusion of the monetary approach to explaining the relationship among monetary policy, domestic credit policy, and the flow of international reserves in the context of BoP. Most components of the balance of payments are directly or indirectly affected by a country's domestic credit and monetary policy. The importance of monetary policy is to restore equilibrium in the money market. Some of the macro-variables such as nominal interest rate, GDP and inflation are affected by contractionary/expansionary monetary policy. And, thus, it has a significant effect on domestic credit as well as the balance of payments of a country.

This is based on the simple idea that the balance of payment surplus or deficit is caused by disequilibrium in the money market. A BoP deficit or surplus represents a transient stock-adjustment process evoked by an initial inequality between actual and desired money stocks. The monetary approach to the BoP assumes fixed exchange rates. Whenever the economy has disequilibrium in the money market, it will result in a payments imbalance but this disequilibrium will be self-correcting through time and there is no need for correcting the disequilibrium through policy adjustments. The main reason for the self-correcting mechanism is that the stable demand function for money relates to money as a stock, not as a flow. When the desired stock is reached, the inflow or outflow of foreign funds stops and so does the balance of payment surplus or deficit. According to *Johnson*, the central point of the monetary approach to BoP is that balance-of-payments deficits or surpluses reflect stock disequilibrium between demand and supply in the market for the money.

Although real factors are not excluded, the demands for the supply of money are the central theoretical relationships used in the analysis of the balance of payments. An increase in domestic credit raises money supply relative to money demand, other things being equal. So the balance of payments must go into deficit to reduce the money supply and restore money market equilibrium. This approach is an explanation of the overall balance of payments. It explains changes in the balance of payments in terms of the demand for and supply of money. According to this approach, *a balance of payments deficit is always and everywhere a monetary phenomenon*. Therefore, it can only be corrected by monetary measures.

10.2.4.2 Monetary approach and fixed exchange rate

The monetary approach implies that a country has no control over its money supply under fixed exchange rates. The money supply is endogenous rather than exogenous or a policy variable. The pursuit of any domestic objective, such as price stability, by altering the domestic component of the monetary base will be frustrated by offsetting changes in the international component through reserve flows. If a country cannot pursue sterilization policies, all the government can do is control the composition of the money supply, not it's level. This implies that under the situation the government cannot determine the level of the monetary base and hence money supply, but only the division of the base into its domestic component, consisting of domestic credit created by the monetary authorities and foreign components, that is, international reserves.

The money demand is defined as positively related to the nations' nominal national income and is assumed to be a stable function in the long run. On the other end of establishing equilibrium in the money market lies the money supply in an economy which is to a large extent given by the monetary authorities of the nation exogenously. The monetary equilibrium condition requires maintaining stability in the money market.

If the disequilibrium results from the inflow of foreign nations' currency it is equivalent to having a balance of payment surplus whereby inflow outturns the outflow of nations' currency abroad. Suppose the increased money demand is not satisfied by the monetary authorities through a rise in the domestic component of the money supply. In that case, the foreign exchange component shall increase creating an imbalance in the external account of the nation. Similarly, suppose the disequilibrium in the money market exists in the form of a higher money supply or a shortage in money demand. In that case, the foreign component of the monetary base flows out of the nation thereby creating a deficit in the nation's balance of payments account. Thus a surplus in the balance of payments accounts results from an excess in the stock of money demand, which is not satisfied by the rise in domestic components of the nation's monetary base, and BoP deficit